DATA ANALYTICS & BUSINESS INTELLIGENCE

TASK 1

Introduction

Data analytics is essential for turning raw business data into useful insights. This project focuses on cleaning sales data, performing exploratory data analysis (EDA), applying SQL queries, and creating dashboards. The goal is to identify key trends, measure performance, and provide clear recommendations that support better business decisions.

Objectives

The main objectives of this project are:

- To clean and preprocess raw sales data by removing duplicates, handling missing values, and standardizing formats.

- To perform exploratory data analysis (EDA) using Excel pivot tables and formulas to identify trends and variations.

- To apply SQL queries for structured analysis, including aggregation and grouping of sales and profit data.

- To design an interactive dashboard in Power BI/Excel with charts, KPI cards, and slicers for better visualization.

- To generate meaningful business insights and provide clear recommendations for management decision-making.

Data Cleaning Steps:

1. Removed Duplicate Records – Identified and deleted duplicate rows to avoid double counting.

2. Handled Missing Values – Replaced or removed missing entries logically to maintain data integrity.

3. Standardized Text Fields – Corrected inconsistent naming (e.g., “east”, “East”, “EAST” → “East”).

4. Corrected Date Formats – Converted all date fields into a uniform format (DD/MM/YYYY).

5. Verified Numerical Columns – Ensured sales, profit, and quantity columns were properly formatted as numbers.

6. Checked for Blank Cells – Filled or removed blank cells where necessary.

PIVOT TABLE:

C24

	A	B	C	D
3	Row Labels	Sum of Profit		
4	East	5015		
5	Furniture	2148.89		
6	Office Supplies	1729.07		
7	Technology	1137.04		
8	North	4797.26		
9	Furniture	1672.63		
10	Office Supplies	1614.34		
11	Technology	1510.29		
12	South	6208.51		
13	Furniture	3005.93		
14	Office Supplies	1709.18		
15	Technology	1493.4		
16	West	5747.26		
17	Furniture	3121.6		
18	Office Supplies	1480.68		
19	Technology	1144.98		
22	Grand Total	21768.03		
23				
24				

EDA:

PIVOT TABLE CHARTS

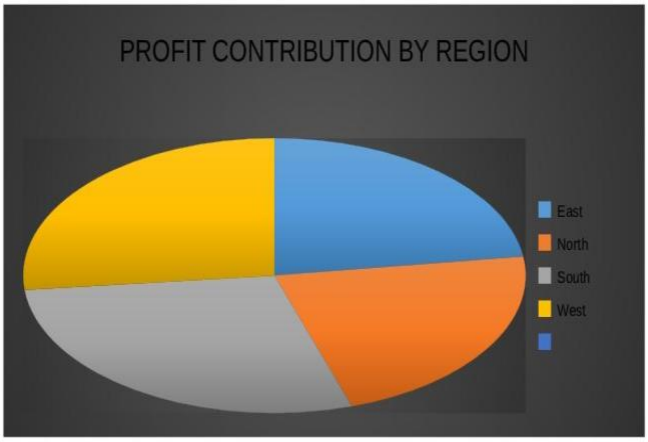
Chart :1

This chart shows total sales across different regions. The South region recorded the highest sales, followed by the East region.

Chart:2

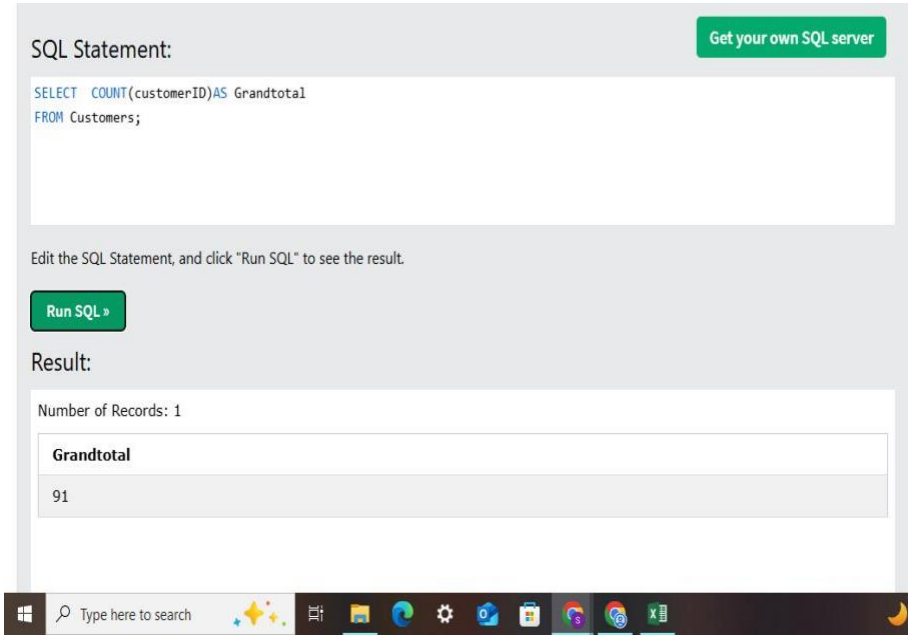
This chart shows how profit changes over time, with clear peaks in the final quarter and dips during high-discount periods

Chart : 3



This chart shows how each region contributes to overall profit. The South region provides the highest share, while some regions show comparatively lower profitability.

SQL QUERIES (DEMOS)



SQL Statement:

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```
SELECT COUNTRY , COUNT(customerID)AS Totalcustomers
FROM Customers
GROUP BY COUNTRY ;
```

Edit the SQL Statement, and click "Run SQL" to see the result.

Run SQL »

Result:

Number of Records: 21

COUNTRY	Totalcustomers
Argentina	3
Austria	2
Belgium	2

Type here to search

Run SQL »

Result:

Number of Records: 21

COUNTRY	Totalcustomers
Argentina	3
Austria	2
Belgium	2
Brazil	9
Canada	3
Denmark	2
Finland	2
France	11

Type here to search

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SQL Statement:

```
SELECT COUNT(OrderID) AS
Total_Orders
FROM Orders;
```

Edit the SQL Statement, and click "Run SQL" to see the result.

Run SQL »

Result:

Number of Records: 1

Total_Orders
196

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SQL Statement:

```
SELECT MAX(CustomerID) AS
MaxCustomer FROM Customers;
```

Edit the SQL Statement, and click "Run SQL" to see the result.

Run SQL »

Result:

Number of Records: 1

MaxCustomer
91

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SQL Statement:

```
SELECT COUNT(OrderID) AS
TotalOrders FROM Orders;
```

Edit the SQL Statement, and click "Run SQL" to see the result.

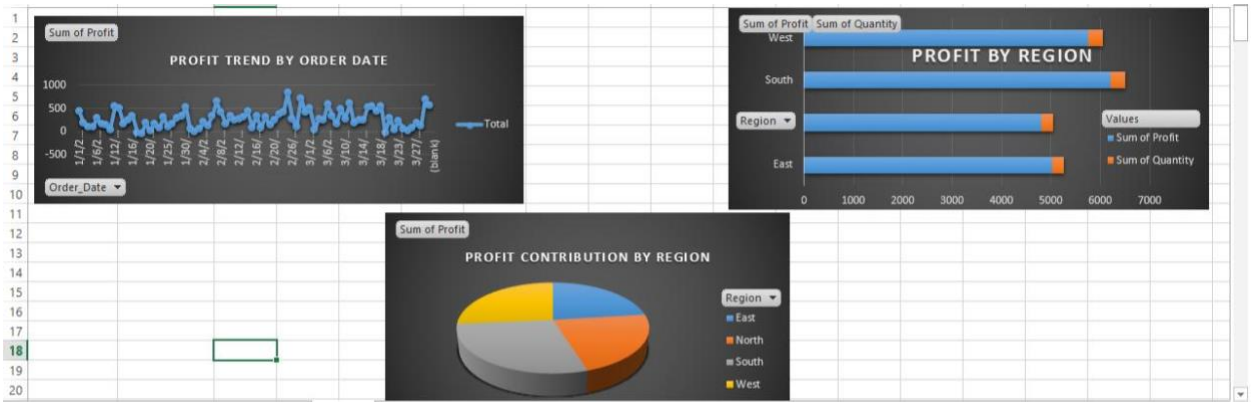
Run SQL »

Result:

Number of Records: 1

TotalOrders
196

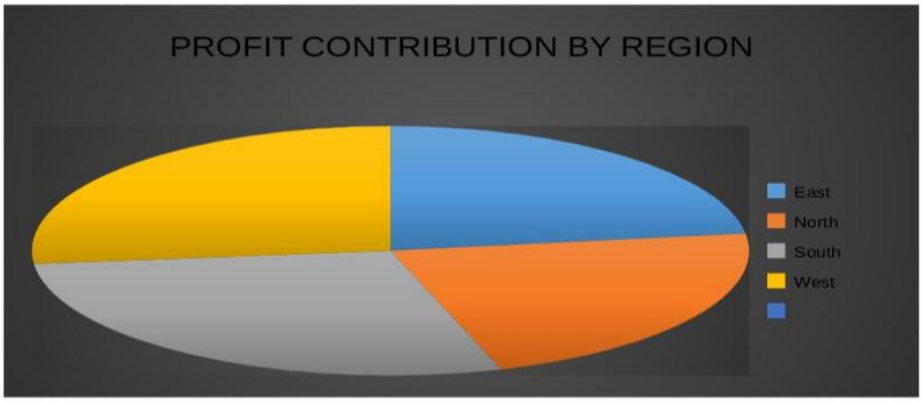
DASHBOARD CREATION:



LINE CHART:

BAR CHART:

PIE CHART;



FILTERS& SLICERS

Business Insights :

1. Which region generates highest sales?

The South region recorded the highest sales, contributing the largest share of overall revenue.

2. Which category is most profitable?

The Technology category generated the highest profit, while Furniture and Office Supplies showed lower margins.

3. Are discounts affecting profit?

Discounts above 20% significantly reduced profit margins, showing that aggressive discounting negatively impacts profitability.

4. Which month performs best?

December emerged as the best-performing month, with peak sales and profit due to year-end demand.

5. Key Recommendations for Management:

- Focus marketing and inventory efforts on the South region and Technology category.

- Strengthen relationships with top customers to secure consistent revenue.

- Optimize discount strategies to balance sales growth with profitability.

- Plan for seasonal demand peaks in December with targeted promotions and stock management.

SUMMARY:

The internship project successfully demonstrated the complete data analytics workflow:

- Data Cleaning: Raw dataset was cleaned and prepared for analysis.

- Exploratory Data Analysis (EDA): Pivot tables and charts revealed key trends in sales, profit, and customer contribution.

- SQL Queries: Aggregation queries were executed to calculate region-wise sales, top customers, and monthly performance.

- Dashboard Creation: An interactive dashboard was built in Excel/Power BI, integrating bar charts, line charts, pie charts, and KPI cards with slicers for dynamic analysis.

- Business Insights:

  - South region generated the highest sales.

  - Technology category was most profitable.

  - Discounts above 20% reduced profit margins.

  - December was the best-performing month.

- Recommendations: Focus on profitable regions and categories, strengthen top customer relationships, optimize discount strategies, and plan for seasonal demand peaks.

Outcome:

This project enhanced skills in Excel, SQL, and BI tools, and delivered a professional, screenshot-ready report that supports management decision-making.